

Data Task UChicago

Sudha Jagatheesh Jayanand

Data Task

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Initial Setup

```
#Core data wrangling & plotting packages  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.4  
## v forcats    1.0.0      v stringr   1.5.1  
## v ggplot2    3.5.1      v tibble    3.2.1  
## v lubridate  1.9.3      v tidyr     1.3.1  
## v purrr      1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
## Warning: package 'janitor' was built under R version 4.3.3
```

```
##  
## Attaching package: 'janitor'  
##  
## The following objects are masked from 'package:stats':  
##  
##      chisq.test, fisher.test
```

```
library(mosaic)
```

```
## Registered S3 method overwritten by 'mosaic':  
##   method                from  
##   fortify.SpatialPolygonsDataFrame ggplot2  
##  
## The 'mosaic' package masks several functions from core packages in order to add  
## additional features. The original behavior of these functions should not be affected by this.
```

```
##
## Attaching package: 'mosaic'
##
## The following object is masked from 'package:Matrix':
##
##     mean
##
## The following objects are masked from 'package:dplyr':
##
##     count, do, tally
##
## The following object is masked from 'package:purrr':
##
##     cross
##
## The following object is masked from 'package:ggplot2':
##
##     stat
##
## The following objects are masked from 'package:stats':
##
##     binom.test, cor, cor.test, cov, fivenum, IQR, median, prop.test,
##     quantile, sd, t.test, var
##
## The following objects are masked from 'package:base':
##
##     max, mean, min, prod, range, sample, sum
```

```
library(psych)
```

```
## Warning: package 'psych' was built under R version 4.3.3
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:mosaic':
##
##     logit, rescale
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(corr)
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2
```

```
library(parameters)
```

```
## Warning: package 'parameters' was built under R version 4.3.3
```

```
library(readr)
library(broom)
library(dplyr)
```

Uploading dataset

```
data <- read_csv("/Users/sudharsanajagatheesh/Desktop/Jackson_2019_Dataset.csv")
```

```
## Rows: 8688 Columns: 25
## -- Column specification -----
## Delimiter: ","
## chr (1): country
## dbl (24): Age, Native, Education, Income, Religion, CT_1, CT_2, CT_3, CT_4, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#Getting a quick look at the data
glimpse(data)
```

```
## Rows: 8,688
## Columns: 25
## $ Age                <dbl> 54, 72, 30, 75, 65, 35, 68, 67, 44, 68, 61, 73, 59, ~
## $ Native             <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4~
## $ Education          <dbl> 2, 1, 2, 1, 1, 2, 1, 1, 2, 2, 2, 1, 1, 1, 1, 1, 1~
## $ Income             <dbl> 3, 2, 1, 2, 2, 5, 3, 3, 1, 2, 1, 2, 1, 2, 2, 1, 1, 2~
## $ Religion           <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 2, 5, 1, 1, 1, 1, 1, 1, 1~
## $ CT_1              <dbl> 4, 6, 4, 4, 4, 5, 7, 4, 4, 4, 5, 7, 7, 6, 5, 7, 7, 6~
## $ CT_2              <dbl> 4, 5, 3, 4, 3, 3, 6, 4, 4, 2, 4, 5, 3, 4, 6, 4, 7, 5~
## $ CT_3              <dbl> 5, 4, 3, 4, 4, 2, 4, 4, 4, 4, 4, 5, 4, 2, 4, 7, 4, 5~
## $ CT_4              <dbl> 5, 5, 4, 4, 4, 5, 7, 4, 4, 3, 5, 6, 7, 6, 5, 7, 7, 6~
## $ CT_5              <dbl> 5, 5, 4, 4, 4, 3, 4, 4, 4, 5, 4, 7, 5, 5, 4, 7, 5, 6~
## $ CT_6              <dbl> 5, 5, 4, 4, 4, 2, 4, 4, 4, 1, 4, 7, 5, 3, 4, 7, 4, 7~
## $ CT_7              <dbl> 4, 5, 3, 4, 4, 4, 6, 4, 4, 4, 4, 7, 7, 4, 4, 7, 4, 5~
## $ CT_8              <dbl> 4, 4, 3, 4, 4, 5, 4, 4, 4, 2, 4, 5, 4, 4, 4, 7, 4, 6~
## $ CT_9              <dbl> 5, 4, 3, 4, 4, 5, 6, 4, 4, 4, 4, 6, 5, 3, 5, 7, 7, 6~
## $ SocietalThreat_1  <dbl> 4, 5, 4, 5, 3, 5, 5, 4, 1, 5, 5, 4, 5, 2, 4, 5, 5, 5~
## $ SocietalThreat_2  <dbl> 4, 3, 4, 5, 2, 3, 3, 4, 1, 5, 3, 5, 2, 4, 4, 3, 5, 5~
## $ SocietalThreat_3  <dbl> 2, 1, 5, 5, 3, 2, 4, 4, 1, 5, 4, 3, 3, 2, 4, 2, 5, 5~
## $ SocietalThreat_4  <dbl> 3, 5, 5, 5, 3, 3, 5, 4, 1, 5, 3, 5, 4, 4, 3, 3, 5, 5~
## $ SocietalThreat_5  <dbl> 3, 5, 5, 5, 2, 4, 5, 4, 1, 2, 4, 5, 3, 2, 4, 4, 5, 5~
## $ SocietalThreat_6  <dbl> 3, 4, 4, 5, 3, 5, 2, 3, 1, 5, 4, 2, 4, 3, 4, 5, 5, 5~
## $ SocietalThreat_7  <dbl> 4, 5, 4, 5, 3, 3, 5, 4, 1, 5, 4, 5, 5, 2, 5, 2, 5, 5~
## $ SocietalThreat_8  <dbl> 2, 4, 5, 5, 3, 4, 5, 4, 5, 5, 3, 3, 5, 4, 4, 2, 5, 5~
## $ SocietalThreat_9  <dbl> 3, 5, 4, 5, 2, 4, 4, 4, 1, 2, 5, 4, 1, 4, 4, 1, 5, 5~
## $ SocietalThreat_10 <dbl> 4, 4, 5, 5, 2, 3, 5, 4, 1, 2, 4, 5, 2, 4, 4, 3, 5, 5~
## $ country           <chr> "USA", "USA", "USA", "USA", "USA", "USA", "USA", "USA", "US~
```

```
#Checking to make sure we have no missing values
summary(data)
```

```
##      Age      Native Education      Income      Religion
## Min.   :18.00   Min.    :4   Min.    :1.000   Min.    :1.000   Min.    : 1.000
## 1st Qu.:25.00   1st Qu.:4   1st Qu.:1.000   1st Qu.:1.000   1st Qu.: 1.000
## Median :35.00   Median :4   Median :1.000   Median :1.000   Median : 1.000
## Mean   :38.04   Mean    :4   Mean    :1.301   Mean    :1.575   Mean    : 2.052
## 3rd Qu.:49.00   3rd Qu.:4   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.: 2.000
## Max.   :95.00   Max.    :4   Max.    :2.000   Max.    :5.000   Max.    :10.000
##                                     NA's    :95
##      CT_1      CT_2      CT_3      CT_4
## Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:4.000   1st Qu.:3.000   1st Qu.:3.000   1st Qu.:4.000
## Median :4.000   Median :4.000   Median :4.000   Median :4.000
## Mean   :4.536   Mean    :4.143   Mean    :3.859   Mean    :4.473
## 3rd Qu.:6.000   3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:5.000
## Max.   :7.000   Max.    :7.000   Max.    :7.000   Max.    :7.000
##
##      CT_5      CT_6      CT_7      CT_8
## Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:4.000   1st Qu.:4.000   1st Qu.:4.000   1st Qu.:4.000
## Median :5.000   Median :5.000   Median :5.000   Median :4.000
## Mean   :4.679   Mean    :4.839   Mean    :4.643   Mean    :4.541
## 3rd Qu.:6.000   3rd Qu.:6.000   3rd Qu.:6.000   3rd Qu.:6.000
## Max.   :7.000   Max.    :7.000   Max.    :7.000   Max.    :7.000
##
##      CT_9      SocietalThreat_1 SocietalThreat_2 SocietalThreat_3
## Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:4.000   1st Qu.:3.000   1st Qu.:3.000   1st Qu.:3.000
## Median :4.000   Median :4.000   Median :4.000   Median :4.000
## Mean   :4.457   Mean    :3.764   Mean    :3.606   Mean    :4.041
## 3rd Qu.:6.000   3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:5.000
## Max.   :7.000   Max.    :5.000   Max.    :5.000   Max.    :5.000
##
##      SocietalThreat_4 SocietalThreat_5 SocietalThreat_6 SocietalThreat_7
## Min.   :1.000   Min.   :1.00   Min.   :1.000   Min.   :1.000
## 1st Qu.:3.000   1st Qu.:3.00   1st Qu.:3.000   1st Qu.:3.000
## Median :4.000   Median :4.00   Median :4.000   Median :4.000
## Mean   :3.759   Mean    :4.04   Mean    :3.778   Mean    :3.876
## 3rd Qu.:5.000   3rd Qu.:5.00   3rd Qu.:5.000   3rd Qu.:5.000
## Max.   :5.000   Max.    :5.00   Max.    :5.000   Max.    :5.000
##
##      SocietalThreat_8 SocietalThreat_9 SocietalThreat_10 country
## Min.   :1.00   Min.   :1.000   Min.   :1.000   Length:8688
## 1st Qu.:3.00   1st Qu.:3.000   1st Qu.:3.000   Class :character
## Median :5.00   Median :3.000   Median :4.000   Mode  :character
## Mean   :4.05   Mean    :3.386   Mean    :3.537
## 3rd Qu.:5.00   3rd Qu.:5.000   3rd Qu.:5.000
## Max.   :5.00   Max.    :5.000   Max.    :5.000
##
```

```
colSums(is.na(data))
```

```
##           Age           Native           Education           Income
##           0             0             0             95
##      Religion           CT_1           CT_2           CT_3
##           0             0             0             0
##      CT_4           CT_5           CT_6           CT_7
##           0             0             0             0
##      CT_8           CT_9 SocietalThreat_1 SocietalThreat_2
##           0             0             0             0
## SocietalThreat_3 SocietalThreat_4 SocietalThreat_5 SocietalThreat_6
##           0             0             0             0
## SocietalThreat_7 SocietalThreat_8 SocietalThreat_9 SocietalThreat_10
##           0             0             0             0
##      country
##           2
```

Cool! The dataset is labelled as mentioned in the *Instructions* file, and there's no missing values, which means we can directly move onto the analysis.

Question 1

The question is essentially asking for a bar plot with variables ranked on descending values of concern for each threat. While a simple bar chart would've provided this information, I was curious to see if there were any differences across threat types (Social, Political, Economic, Environmental, Health), so I colour-coded the `SocietalThreat` variables based on general threat domain, and plotted the bar chart with names mapped from the variable codes. Confidence intervals were also plotted to show the certainty of these scores.

```
long_threats <- data %>%
  pivot_longer(cols = starts_with("SocietalThreat"),
    names_to = "Threat",
    values_to = "Concern") %>%
  mutate(
    Threat = recode(Threat,
      SocietalThreat_1 = "Natural Disaster",
      SocietalThreat_2 = "Famine",
      SocietalThreat_3 = "Pollution",
      SocietalThreat_4 = "Terrorist Attack",
      SocietalThreat_5 = "Crime",
      SocietalThreat_6 = "Discrimination",
      SocietalThreat_7 = "National Debt",
      SocietalThreat_8 = "COVID-19",
      SocietalThreat_9 = "Refugee Influx",
      SocietalThreat_10 = "Illegal Immigration"
    ),
    Category = case_when(
      Threat %in% c("Natural Disaster", "Famine", "Pollution") ~ "Environmental",
      Threat %in% c("Terrorist Attack", "Crime", "Discrimination") ~ "Social",
      Threat %in% c("Refugee Influx", "Illegal Immigration") ~ "Migration",
```

```

    Threat == "COVID-19" ~ "Health",
    Threat == "National Debt" ~ "Economic"
  )
)

#Summary stats
threat_summary <- long_threats %>%
  group_by(Threat, Category) %>%
  summarise(
    mean_concern = mean(Concern, na.rm = TRUE),
    sd = sd(Concern, na.rm = TRUE),
    n = sum(!is.na(Concern)),
    se = sd / sqrt(n),
    ci_low = mean_concern - 1.96 * se,
    ci_high = mean_concern + 1.96 * se
  ) %>%
  ungroup() %>%
  arrange(desc(mean_concern))

```

`summarise()` has grouped output by 'Threat'. You can override using the
`.groups` argument.

```

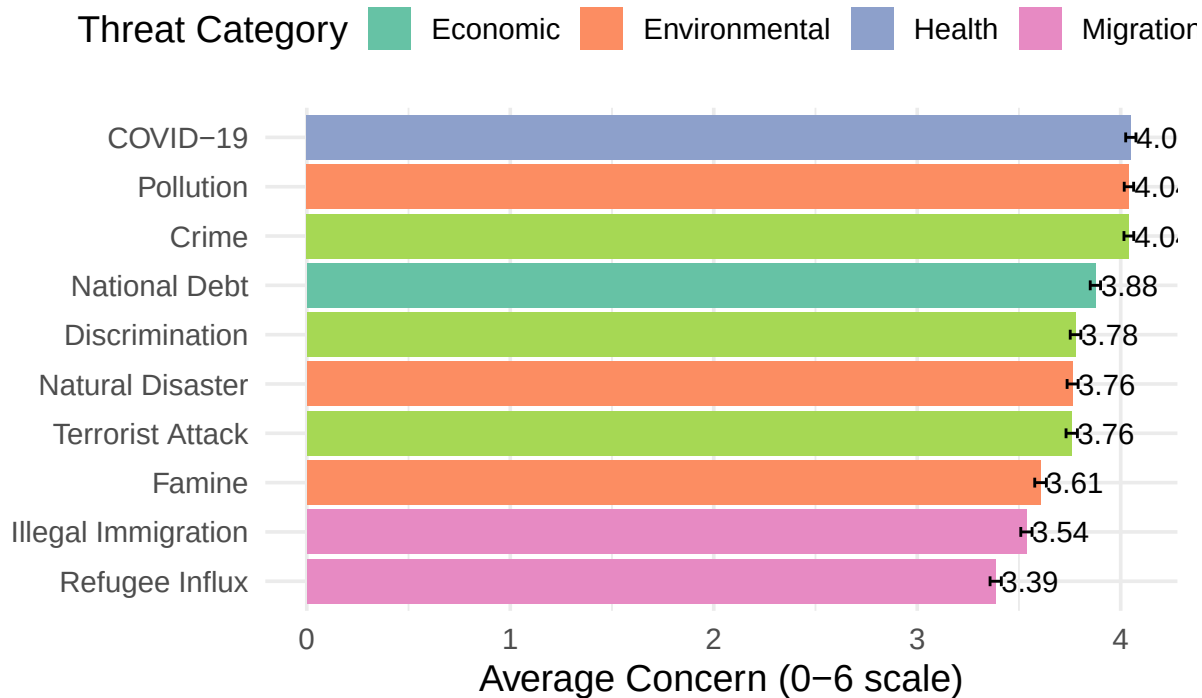
library(ggplot2)

ggplot(threat_summary, aes(x = reorder(Threat, mean_concern), y = mean_concern, fill = Category)) +
  geom_col() +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2, color = "black") +
  geom_text(aes(label = round(mean_concern, 2)), hjust = -0.1, size = 4) +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Which Threats Worry People the Most?",
    subtitle = "Average concern across all countries, grouped by threat category",
    x = "",
    y = "Average Concern (0-6 scale)",
    fill = "Threat Category"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")

```

Which Threats Worry People the Most?

Average concern across all countries, grouped by threat



The output seems interesting! Covid-19 has the highest average concern, but it isn't far from Pollution and Crime, which together make up the top three threat categories. Interestingly, however, Illegal Immigration and Refugee Influx both score the lowest, which suggests that on average, issues on immigration seem to have lower levels of threat concern.

Question 2

According to Jackson (2019), the desire for cultural tightness scale was validated in previous studies using Chronbach's alpha, factor analysis and measurement in variance across cultures (Gelfand et al., 2011). I decided to implement a *Chronbach's Alpha* test, but decided to also employ a *McDonald's Omega* test. The decision was taken with the understanding that equal item loadings (as assumed in Chronbach's Alpha!) may not necessarily be true, and for a more robust estimate, McDonald's Omega fares better. Essentially, McDonald's omega uses actual factor loadings to weigh the items, giving us a more accurate (albeit marginally different) result.

The two tests demonstrated high internal consistency (*Cronbach's* = .85), with an omega total of .87 and omega general of .72.

Similar to Gelfand et al., I was curious to see what an exploratory factor analysis (EFA) would yield, so I implemented the test to our dataset as it's likely results can vary due to context and other confounds. The results supported a unidimensional structure, with all items loading .50 on a single latent factor. While a bifactor model with three subfactors showed improved fit, the additional group variance was minimal (*omega group range* = .07-.24), and the general factor remained dominant. Essentially, the data is testing a single construct (cultural tightness), which is great!

Chronbach's Alpha

```
ct_items <- data %>%  
  select(starts_with("CT_"))
```

```
library(psych)
```

```
alpha(ct_items)
```

```
##  
## Reliability analysis  
## Call: alpha(x = ct_items)  
##  
##      raw_alpha std.alpha G6(smc) average_r S/N      ase mean   sd median_r  
##      0.85      0.85      0.84      0.39 5.8 0.0024  4.5 1.1      0.38  
##  
##      95% confidence boundaries  
##              lower alpha upper  
## Feldt      0.85  0.85  0.86  
## Duhachek   0.85  0.85  0.86  
##  
## Reliability if an item is dropped:  
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se  var.r med.r  
## CT_1      0.85      0.85      0.83      0.41 5.5  0.0025 0.0055  0.39  
## CT_2      0.85      0.85      0.84      0.41 5.5  0.0025 0.0053  0.40  
## CT_3      0.84      0.84      0.83      0.40 5.4  0.0025 0.0060  0.39  
## CT_4      0.84      0.84      0.82      0.39 5.1  0.0026 0.0068  0.38  
## CT_5      0.83      0.83      0.82      0.38 4.9  0.0028 0.0046  0.38  
## CT_6      0.83      0.83      0.82      0.38 4.9  0.0027 0.0047  0.38  
## CT_7      0.83      0.83      0.82      0.38 4.8  0.0028 0.0054  0.37  
## CT_8      0.84      0.84      0.82      0.39 5.1  0.0026 0.0065  0.38  
## CT_9      0.83      0.83      0.82      0.38 4.9  0.0027 0.0063  0.36  
##  
## Item statistics  
##      n raw.r std.r r.cor r.drop mean  sd  
## CT_1 8688 0.60 0.60 0.52 0.48 4.5 1.6  
## CT_2 8688 0.59 0.59 0.51 0.47 4.1 1.6  
## CT_3 8688 0.61 0.61 0.53 0.49 3.9 1.7  
## CT_4 8688 0.67 0.68 0.62 0.57 4.5 1.6  
## CT_5 8688 0.74 0.73 0.69 0.63 4.7 1.8  
## CT_6 8688 0.73 0.72 0.69 0.63 4.8 1.7  
## CT_7 8688 0.74 0.74 0.71 0.66 4.6 1.6  
## CT_8 8688 0.68 0.68 0.63 0.58 4.5 1.6  
## CT_9 8688 0.73 0.73 0.69 0.64 4.5 1.6  
##  
## Non missing response frequency for each item  
##      1 2 3 4 5 6 7 miss  
## CT_1 0.06 0.03 0.09 0.33 0.22 0.10 0.16 0  
## CT_2 0.08 0.06 0.14 0.35 0.19 0.07 0.10 0  
## CT_3 0.14 0.07 0.13 0.35 0.16 0.06 0.09 0  
## CT_4 0.06 0.04 0.10 0.35 0.21 0.08 0.15 0  
## CT_5 0.09 0.05 0.09 0.23 0.21 0.12 0.22 0
```

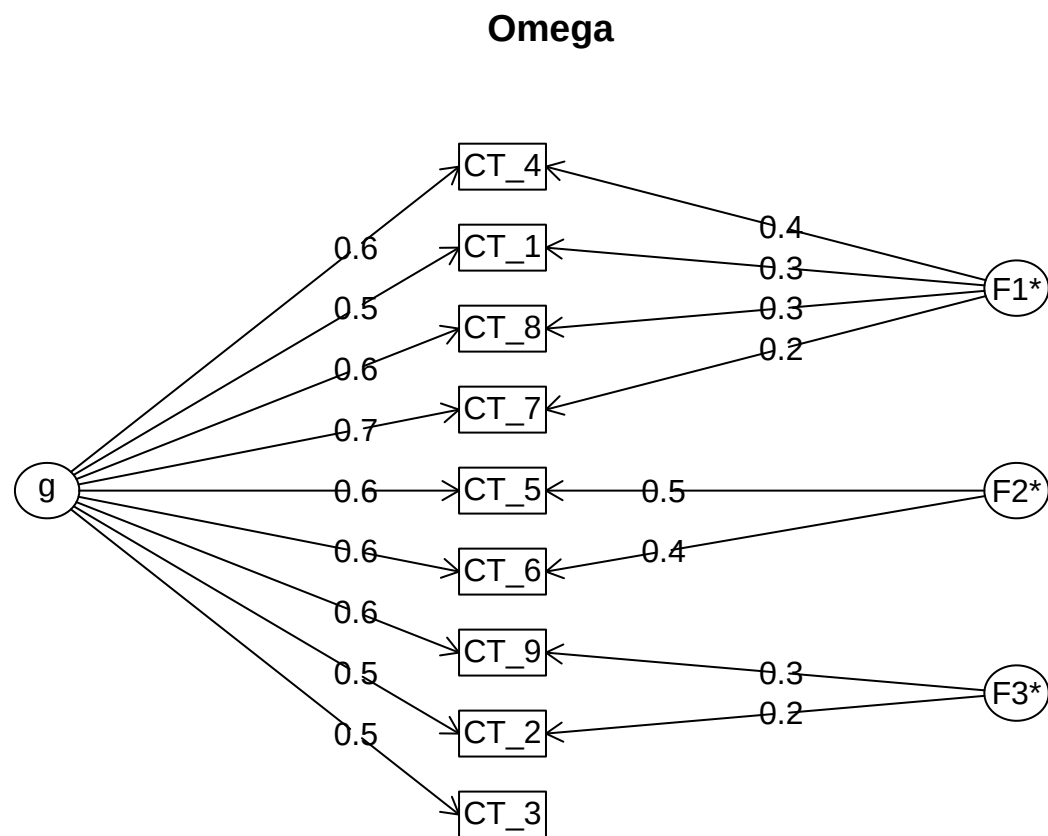


```
## CT_6 0.05 0.04 0.08 0.26 0.21 0.13 0.24 0
## CT_7 0.05 0.04 0.10 0.31 0.22 0.09 0.19 0
## CT_8 0.05 0.04 0.11 0.32 0.21 0.10 0.16 0
## CT_9 0.07 0.04 0.10 0.33 0.21 0.10 0.15 0
```

McDonald's Omega

```
library(parameters)
omega(ct_items)
```

```
## Loading required namespace: GPArotation
```



```
## Omega
## Call: omegah(m = m, nfactors = nfactors, fm = fm, key = key, flip = flip,
##   digits = digits, title = title, sl = sl, labels = labels,
##   plot = plot, n.obs = n.obs, rotate = rotate, Phi = Phi, option = option,
##   covar = covar)
## Alpha:          0.85
## G.6:            0.84
## Omega Hierarchical: 0.72
## Omega H asymptotic: 0.83
## Omega Total      0.87
##
```

```

## Schmid Leiman Factor loadings greater than 0.2
##      g  F1*  F2*  F3*  h2  h2  u2  p2  com
## CT_1 0.49  0.31      0.33 0.33 0.67 0.71 1.70
## CT_2 0.47      0.23 0.29 0.29 0.71 0.76 1.63
## CT_3 0.49      0.30 0.30 0.70 0.80 1.51
## CT_4 0.58  0.37      0.49 0.49 0.51 0.70 1.75
## CT_5 0.65      0.48      0.66 0.66 0.34 0.64 1.87
## CT_6 0.63      0.39      0.56 0.56 0.44 0.72 1.71
## CT_7 0.65  0.21      0.51 0.51 0.49 0.83 1.42
## CT_8 0.58  0.29      0.44 0.44 0.56 0.75 1.67
## CT_9 0.64      0.29 0.53 0.53 0.47 0.78 1.57
##
## With Sums of squares of:
##      g  F1*  F2*  F3*  h2
## 3.03 0.40 0.48 0.19 2.00
##
## general/max 1.51  max/min = 10.34
## mean percent general = 0.74  with sd = 0.06 and cv of 0.08
## Explained Common Variance of the general factor = 0.74
##
## The degrees of freedom are 12  and the fit is 0.01
## The number of observations was 8688  with Chi Square = 71.26  with prob < 1.9e-10
## The root mean square of the residuals is 0.01
## The df corrected root mean square of the residuals is 0.01
## RMSEA index = 0.024  and the 90 % confidence intervals are 0.019 0.029
## BIC = -37.58
##
## Compare this with the adequacy of just a general factor and no group factors
## The degrees of freedom for just the general factor are 27  and the fit is 0.18
## The number of observations was 8688  with Chi Square = 1540.37  with prob < 2.3e-308
## The root mean square of the residuals is 0.07
## The df corrected root mean square of the residuals is 0.08
##
## RMSEA index = 0.08  and the 90 % confidence intervals are 0.077 0.084
## BIC = 1295.48
##
## Measures of factor score adequacy
##
##      g  F1*  F2*  F3*
## Correlation of scores with factors 0.86 0.50 0.58 0.44
## Multiple R square of scores with factors 0.73 0.25 0.34 0.19
## Minimum correlation of factor score estimates 0.47 -0.50 -0.32 -0.62
##
## Total, General and Subset omega for each subset
##
##      g  F1*  F2*  F3*
## Omega total for total scores and subscales 0.87 0.74 0.75 0.59
## Omega general for total scores and subscales 0.72 0.59 0.52 0.50
## Omega group for total scores and subscales 0.07 0.15 0.24 0.10

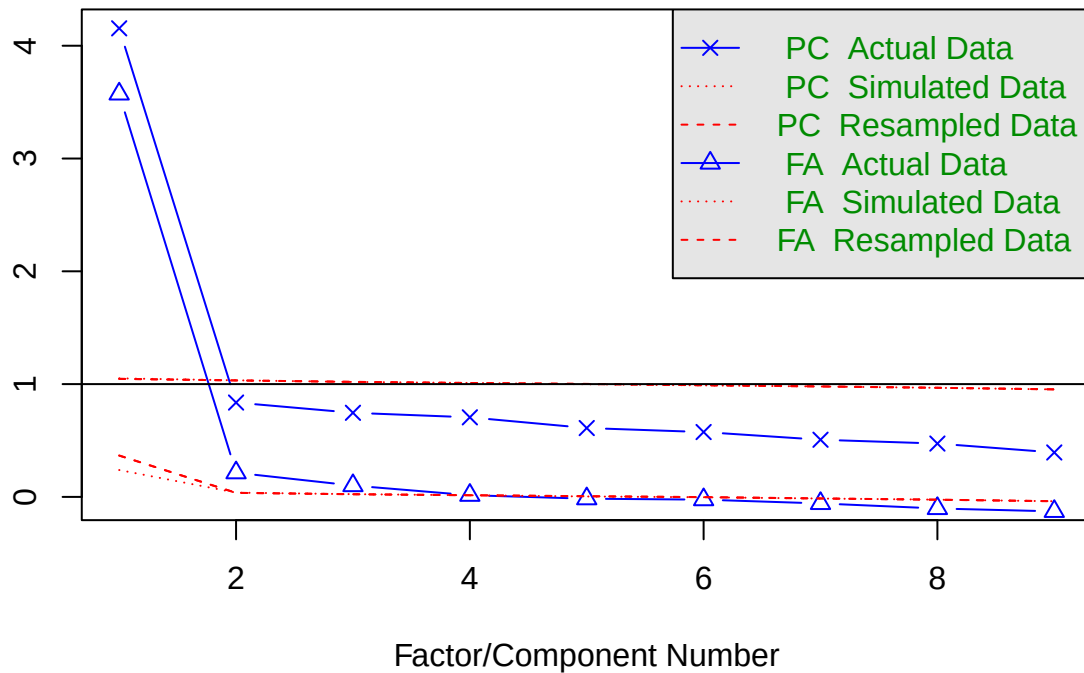
```

Exploratory Factor Analysis

```
fa.parallel(ct_items) # Scree plot for number of factors
```

eigenvalues of principal components and factor analysis

Parallel Analysis Scree Plots



Parallel analysis suggests that the number of factors = 3 and the number of components = 1

```
fa(ct_items, nfactors = 1, rotate = "none") # Basic 1-factor solution
```

```
## Factor Analysis using method = minres
## Call: fa(r = ct_items, nfactors = 1, rotate = "none")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1  h2  u2 com
## CT_1 0.52 0.27 0.73  1
## CT_2 0.51 0.26 0.74  1
## CT_3 0.53 0.28 0.72  1
## CT_4 0.62 0.38 0.62  1
## CT_5 0.70 0.49 0.51  1
## CT_6 0.70 0.49 0.51  1
## CT_7 0.72 0.52 0.48  1
## CT_8 0.63 0.40 0.60  1
## CT_9 0.70 0.49 0.51  1
##
##              MR1
## SS loadings  3.57
## Proportion Var 0.40
##
## Mean item complexity = 1
## Test of the hypothesis that 1 factor is sufficient.
##
## df null model = 36 with the objective function = 2.81 with Chi Square = 24357.91
## df of the model are 27 and the objective function was 0.12
```

```
##
## The root mean square of the residuals (RMSR) is 0.03
## The df corrected root mean square of the residuals is 0.04
##
## The harmonic n.obs is 8688 with the empirical chi square 759.77 with prob < 1.1e-142
## The total n.obs was 8688 with Likelihood Chi Square = 1073.91 with prob < 5.1e-209
##
## Tucker Lewis Index of factoring reliability = 0.943
## RMSEA index = 0.067 and the 90 % confidence intervals are 0.063 0.07
## BIC = 829.03
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##
## Correlation of (regression) scores with factors MR1 0.93
## Multiple R square of scores with factors 0.86
## Minimum correlation of possible factor scores 0.73
```

Question 3

Mean CT Scores

```
#mean CT score
data <- data %>%
  mutate(CT_mean = rowMeans(select(., starts_with("CT_")), na.rm = TRUE))

#threat variable list
threat_vars <- names(data)[grepl("^SocietalThreat_", names(data))]

#Compute correlations
library(broom)

cor_table <- map_df(threat_vars, function(var) {
  test <- cor.test(data[[var]], data$CT_mean)
  tibble(
    Threat = var,
    r = round(test$estimate, 2),
    p = signif(test$p.value, 3),
    n = sum(complete.cases(data[[var]], data$CT_mean))
  )
})

#Mapping the names back to variables
cor_table$Threat <- recode(cor_table$Threat,
  SocietalThreat_1 = "Natural Disaster",
  SocietalThreat_2 = "Famine",
  SocietalThreat_3 = "Pollution",
  SocietalThreat_4 = "Terrorist Attack",
  SocietalThreat_5 = "Crime",
  SocietalThreat_6 = "Discrimination",
  SocietalThreat_7 = "National Debt",
  SocietalThreat_8 = "COVID-19",
```

```

SocietalThreat_9 = "Refugee Influx",
SocietalThreat_10 = "Illegal Immigration"
)

print(cor_table)

## # A tibble: 10 x 4
##   Threat                r          p      n
##   <chr>          <dbl>    <dbl> <int>
## 1 Natural Disaster    0.09 1.66e-18  8688
## 2 Famine              0.1  1.49e-20  8688
## 3 Pollution           0.11 3.29e-25  8688
## 4 Terrorist Attack    0.05 2.13e- 6  8688
## 5 Crime               0.2  1.48e-76  8688
## 6 Discrimination      0.08 4.66e-13  8688
## 7 National Debt       0.16 1.27e-52  8688
## 8 COVID-19            0.11 2.39e-23  8688
## 9 Refugee Influx      0.18 1.28e-63  8688
## 10 Illegal Immigration 0.17 2.09e-58  8688

```

Correlation Table

A Pearson's correlation test confirmed all correlations were statistically significant but weak in magnitude, with none exceeding $r = .20$. The strongest associations were with **Crime** ($r = .20$), **Refugee Influx** ($r = .18$), and **Illegal Immigration** ($r = .17$). These small values imply that tightness attitudes are influenced by more than just perceived threat.

```

cor_table <- cor_table %>%
  arrange(desc(r)) %>%
  mutate(Threat = factor(Threat, levels = rev(Threat)))

ggplot(cor_table, aes(x = "", y = Threat, fill = r)) +
  geom_tile() +
  geom_text(aes(label = round(r, 2)), color = "black", size = 4) +
  scale_fill_gradient2(low = "green", mid = "white", high = "red", midpoint = 0) +
  labs(
    title = "Correlation Between Threat Concern and Cultural Tightness",
    x = "",
    y = "",
    fill = "Pearson's r"
  ) +
  theme_minimal(base_size = 13) +
  theme(axis.text.x = element_blank())

```

Correlation Between Threat Concern and Cultural Tightness



Question 4

Regression (without controls)

I conducted a simple linear regression to test whether concern about a surge in COVID-19 cases predicts desire for cultural tightness.

The results showed a statistically significant association, $\beta = 0.10$, $SE = 0.0099$, $t(8686) = 9.98$, $p < .001$. Essentially, COVID-19 concern positively predicts the need for cultural tightness. The model, however, explained only a small proportion of variance in tightness scores ($R^2 = 0.011$), indicating that while the relationship is robust due to the large sample size, the effect size is small.

This finding is consistent with the hypothesis that societal threat increases demand for stronger social norms (but, it also shows that COVID-specific concern accounts for only a small portion of individual variation in tightness attitudes!). The result aligns with Jackson et al. (2019), who theorized that ecological threats can trigger tightening responses. There is a need to consider moderators such as cultural background, institutional trust, or baseline norm strength, though, and I decided to include controls given in the dataset to test if the relationship significantly changes upon adding them.

```
model <- lm(CT_mean ~ SocietalThreat_8, data = data)
summary(model)
```

```
##
## Call:
## lm(formula = CT_mean ~ SocietalThreat_8, data = data)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -3.5573 -0.5694 -0.0382  0.6650  2.8385
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.06255    0.04186   97.058 <2e-16 ***
## SocietalThreat_8  0.09894    0.00991    9.984 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.108 on 8686 degrees of freedom
## Multiple R-squared:  0.01135,    Adjusted R-squared:  0.01123
## F-statistic: 99.68 on 1 and 8686 DF,  p-value: < 2.2e-16
```

Regression (with Controls)

In selecting controls, I chose not to include **Religion**, since the coding in this dataset made it unclear whether the variable captured religious affiliation or degree of religiosity (e.g., how important religion is in daily life). Without this distinction, including the variable would have introduced some uncertainty around interpretation. **Income**, **Age**, and **Education** were straightforward, with **Education** being a binary variable, and **Income** being categorical (higher value = higher income bracket).

Next, I ran a multiple linear regression to examine whether concern about a rise in COVID-19 cases predicted participants' desire for cultural tightness, while controlling for age, education, and income. COVID-19 concern was a statistically significant positive predictor ($\beta = 0.097$, $SE = 0.0099$, $t = 9.86$, $p < .001$), even after accounting for key demographic factors.

This model, similar to the one above, explained only a small proportion of variance (about 3%; $R^2 = 0.032$), suggesting that while the association is robust due to the large sample size, its substantive impact is limited. **Age** had a small positive effect on the desire for cultural tightness ($\beta = 0.011$, $p < .001$), and **Education** had a small negative effect ($\beta = -0.068$, $p = .012$). Essentially, individuals who were older or uneducated were more likely to express a desire for cultural tightness. **Income** was not a significant predictor ($p = 0.22$).

Overall, these results are consistent with the theory that ecological threat increases support for stronger social norms. However, the small effect sizes also point to the limited explanatory power of COVID-specific concern alone.

```
model <- lm(CT_mean ~ SocietalThreat_8 + Age + Education + Income, data = data)
summary(model)
```

```
##
## Call:
## lm(formula = CT_mean ~ SocietalThreat_8 + Age + Education + Income,
##     data = data)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -3.8443 -0.6149 -0.0737  0.6677  3.0463
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.735258    0.0612985   60.940 <2e-16 ***
## SocietalThreat_8  0.0974038    0.0098797    9.859 <2e-16 ***
## Age              0.0105577    0.0008064   13.092 <2e-16 ***
```

```
## Education      -0.0680739  0.0271363  -2.509   0.0121 *
## Income         0.0141134  0.0116034   1.216   0.2239
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.096 on 8588 degrees of freedom
## (95 observations deleted due to missingness)
## Multiple R-squared:  0.0323, Adjusted R-squared:  0.03185
## F-statistic: 71.66 on 4 and 8588 DF,  p-value: < 2.2e-16
```

Question 5

Similar to the methodology implemented in Jackson et al. (2019), I decided to pick three countries based on low, moderate, and high desire for cultural tightness. This would help us to see if any patterns emerge in the direction and strength of the effect sizes. So, I plot `mean_CT` scores and visualise them in a table, picking countries with the highest, moderate, and lowest scores for the visualisation.

Plotting CT scores

```
data <- data %>%
  mutate(CT_mean = rowMeans(select(., starts_with("CT_")), na.rm = TRUE))

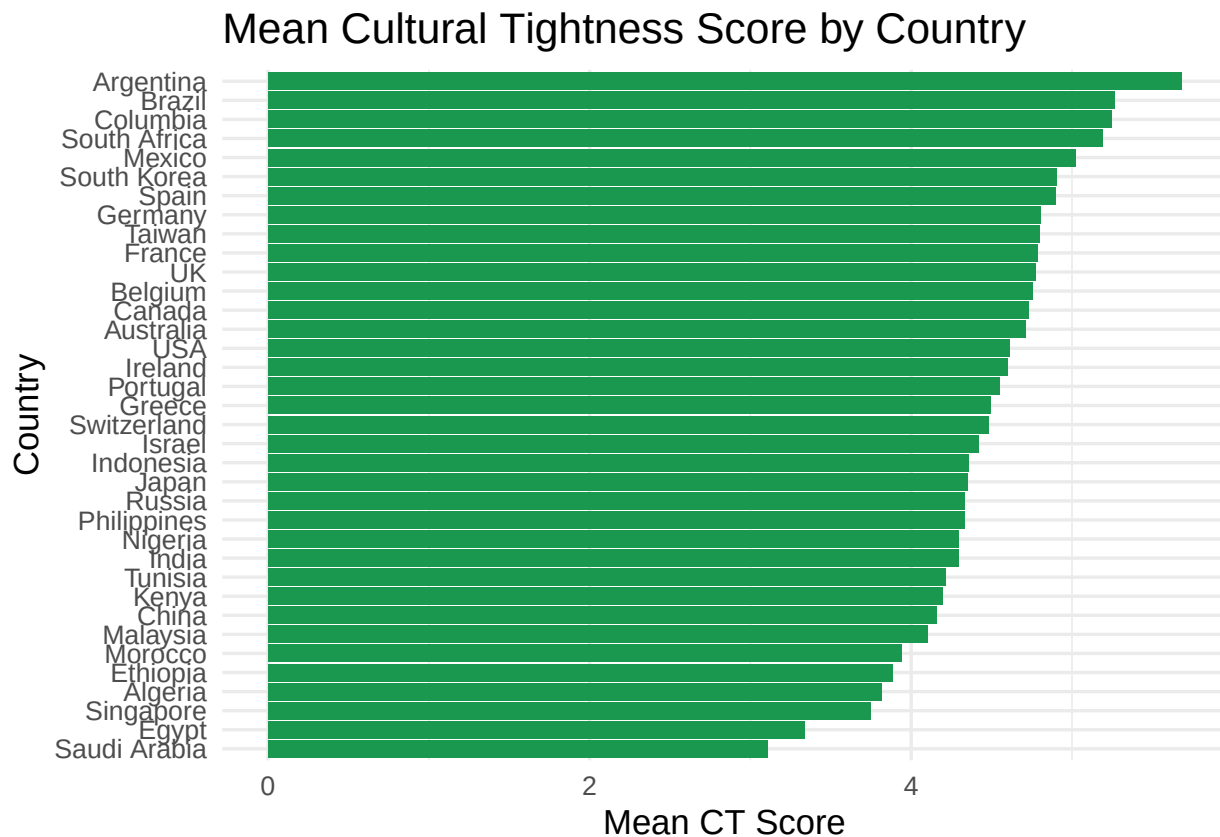
# Get mean tightness by country
ct_by_country <- data %>%
  group_by(country) %>%
  summarise(
    mean_CT = mean(CT_mean, na.rm = TRUE),
    n = n()
  ) %>%
  arrange(desc(mean_CT))

print(ct_by_country)
```

```
## # A tibble: 39 x 3
##   country      mean_CT      n
##   <chr>      <dbl> <int>
## 1 Argentina    5.68   208
## 2 Brazil       5.26   208
## 3 Columbia     5.24   208
## 4 South Africa 5.19   208
## 5 Mexico       5.02   208
## 6 South Korea  4.90   361
## 7 Spain        4.90   208
## 8 Germany      4.80   201
## 9 Taiwan       4.80   361
## 10 France      4.78   208
## # i 29 more rows
```


Selecting countries

```
data %>%
  mutate(CT_mean = rowMeans(select(., starts_with("CT_")), na.rm = TRUE)) %>%
  group_by(country) %>%
  summarise(mean_CT = mean(CT_mean, na.rm = TRUE), n = n()) %>%
  filter(n >= 100) %>% # Optional: filter for countries with decent sample sizes
  ggplot(aes(x = reorder(country, mean_CT), y = mean_CT)) +
  geom_col(fill = "#1a9850") +
  coord_flip() +
  labs(
    title = "Mean Cultural Tightness Score by Country",
    x = "Country",
    y = "Mean CT Score"
  ) +
  theme_minimal(base_size = 13)
```



Final Dot plot

```
plot_data <- data %>%
  filter(country %in% c("Argentina", "Greece", "Saudi Arabia")) %>%
  mutate(CT_mean = rowMeans(select(., starts_with("CT_")), na.rm = TRUE))

results <- plot_data %>%
```

```

group_by(country) %>%
do(tidy(lm(CT_mean ~ SocietalThreat_8, data = .))) %>%
filter(term == "SocietalThreat_8") %>%
select(country, estimate, std.error, p.value) %>%
mutate(
  estimate = round(estimate, 3),
  p.value = signif(p.value, 3)
)

ggplot(results, aes(x = reorder(country, estimate), y = estimate)) +
  geom_point(size = 4, color = "#D55E00") +
  geom_errorbar(aes(ymin = estimate - std.error, ymax = estimate + std.error), width = 0.1, color = "#D55E00") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +
  coord_flip() +
  labs(
    title = "Effect of COVID-19 Concern on Countries with Low, Moderate, and High Desire for Cultural T",
    subtitle = "Regression coefficients ( ) with standard errors for each country",
    x = "Country",
    y = "Regression Coefficient ( )"
  ) +
  theme_minimal(base_size = 13)

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression coefficients ( ) with standard errors for
## each country' in 'mbcsToSbcs': dot substituted for <ce>

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression coefficients ( ) with standard errors for
## each country' in 'mbcsToSbcs': dot substituted for <b2>

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <ce>

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <b2>

```

```

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <ce>

```

```

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <b2>

```

```

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression coefficients ( ) with standard errors for
## each country' in 'mbcsToSbcs': dot substituted for <ce>

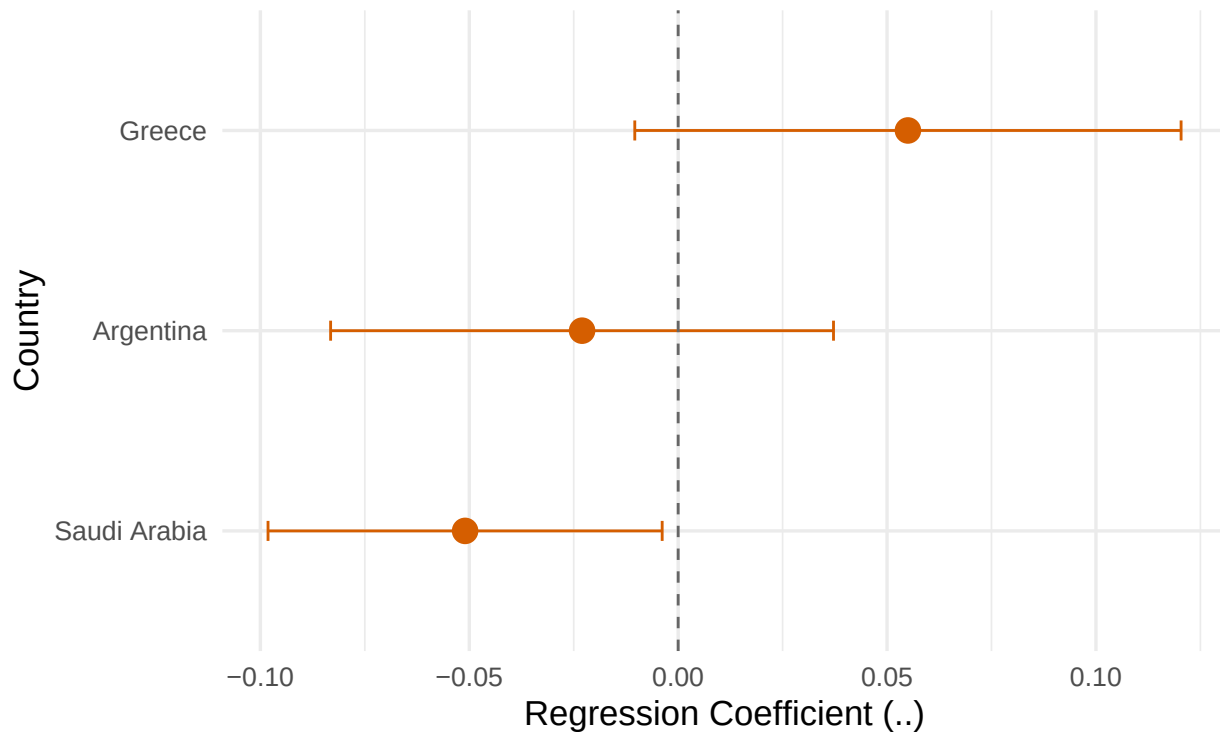
```

```

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression coefficients ( ) with standard errors for
## each country' in 'mbcsToSbcs': dot substituted for <b2>

```

Effect of COVID-19 Concern on Countries with Low, M Regression coefficients (..) with standard errors for each country



This is a very interesting output. As initial theory would suggest, one could expect to see some kind of pattern based on the strength of desire for cultural tightness and Covid-19 concern, but the betas for all three show no such output. This would naturally lead us to conclude that there's no significant differences based on cultural tightness, but what if the three countries I selected just had the one unlucky outlier? So, I decided to check the statistical significance and directionality of the relationship for each country in the dataset.

Plotting regression outputs

```
country_lm_results <- data %>%
  mutate(CT_mean = rowMeans(select(., starts_with("CT_")), na.rm = TRUE)) %>%
  group_by(country) %>%
  do(tidy(lm(CT_mean ~ SocietalThreat_8, data = .))) %>%
  filter(term == "SocietalThreat_8") %>%
  mutate(
    estimate = round(estimate, 3),
    p.value = signif(p.value, 3),
    sig = ifelse(p.value < 0.05, "Yes", "No")
  ) %>%
  arrange(p.value)

print(country_lm_results)
```

```
## # A tibble: 39 x 7
## # Groups:   country [39]
```

```
##      country      term      estimate std.error statistic   p.value sig
##      <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl> <chr>
##  1 Mexico      SocietalThreat_8  0.31    0.0685     4.53 0.00001  Yes
##  2 China       SocietalThreat_8  0.155   0.0383     4.03 0.0000707 Yes
##  3 South Korea SocietalThreat_8  0.19    0.0533     3.55 0.000431  Yes
##  4 France      SocietalThreat_8  0.206   0.0645     3.19 0.00165   Yes
##  5 Benin       SocietalThreat_8 -0.552   0.191     -2.89 0.016     Yes
##  6 India       SocietalThreat_8  0.091   0.0383     2.37 0.018     Yes
##  7 Ireland     SocietalThreat_8  0.121   0.0525     2.30 0.0227    Yes
##  8 Portugal    SocietalThreat_8  0.148   0.0684     2.17 0.0313    Yes
##  9 Taiwan      SocietalThreat_8  0.08    0.0381     2.11 0.0354    Yes
## 10 South Africa SocietalThreat_8  0.127   0.0602     2.11 0.0361    Yes
## # i 29 more rows
```

```
country_counts <- data %>% count(country)
large_countries <- country_counts %>% filter(n >= 100) %>% pull(country)

filtered_results <- country_lm_results %>%
  filter(country %in% large_countries)
```

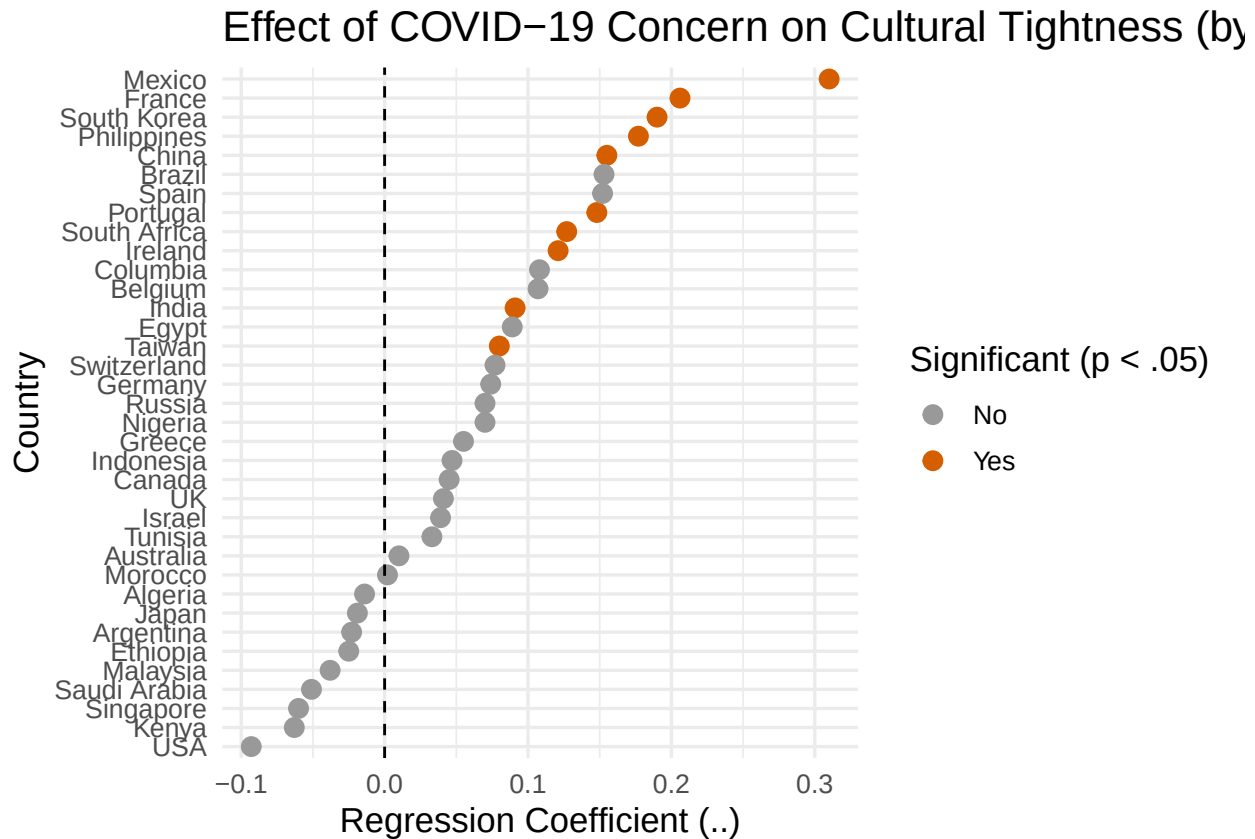
```
ggplot(filtered_results, aes(x = reorder(country, estimate), y = estimate, color = sig)) +
  geom_point(size = 3) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  coord_flip() +
  labs(
    title = "Effect of COVID-19 Concern on Cultural Tightness (by Country)",
    x = "Country",
    y = "Regression Coefficient ( )",
    color = "Significant (p < .05)"
  ) +
  scale_color_manual(values = c("Yes" = "#D55E00", "No" = "gray60")) +
  theme_minimal(base_size = 13)
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <ce>
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <b2>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <ce>
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Regression Coefficient ( )' in 'mbcsToSbcs': dot
## substituted for <b2>
```



This seems to confirm my initial hunch. There's only **10 countries** for which the relationship is statistically significant, so the association is not universal, and seems to be heavily dependent on national contexts. Most of them do not have a significant effect and there is no consistency in the direction of this effect either.

Essentially, this relationship seems to be context-specific in countries; there's other country-specific psychological/social variables at play here that need to be explored.

What does this mean for the regression I conducted previously? Well, countries like Mexico and France have strong positive slopes ($> .25$), and they pull the overall regression up. Basically, this seems to be a classic case of an ecological fallacy, and we've discovered that a relationship at the aggregate level does not mean it holds at the individual level in every subgroup. (viz., country).

References:

- Gelfand, M.J. (2011). Differences Between Tight and Loose Cultures: A 33-Nation Study. *Science* (New York, N.Y.). 332. 1100-4. doi:https://doi.org/10.1126/science.1197754.
- Jackson, J.C., van Egmond, M., Choi, V.K., Ember, C.R., Halberstadt, J., Balanovic, J., Basker, I.N., Boehnke, K., Buki, N., Fischer, R., Fulop, M., Fulmer, A., Homan, A.C., van Kleef, G.A., Kreemers, L., Schei, V., Szabo, E., Ward, C. and Gelfand, M.J. (2019). Ecological and cultural

factors underlying the global distribution of prejudice. *PLOS ONE*, 14(9), p.e0221953. doi:<https://doi.org/10.1371/journal.pone.0221953>.